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%QUESTION 1(b)
%Solving analytically
syms t
syms v(t)
ode = diff(v,t) == g - (c/m)*v;           % dv/dt = g - (c/m)*v
vSol = dsolve(ode, v(0)==v0);
vSol = simplify(vSol) ;                    % display symbolic expression
disp(vSol)

49/2 - (49*exp(-(2*t)/5))/2


%Using ODE45 solver to solve an equation: m(dv/dt) + cv = mg;
v0 = 0;
tRange = 0:1:10;
m = 5.0;
c = 2.0;
g = 9.8;
%Differential equation
V_t = @(t, v) (m * g - c * v) / m;


% Solve the ODE using ode45
[t, v] = ode45(V_t, tRange, v0);
%Graphing the result,
plot(t, v, 'Marker', 'o');
xlabel('Time (s)');
ylabel('Velocity (m/s)');
title('Velocity vs Time');
grid on
%Tabulating the results
results = table(t, v, 'VariableNames', {'Time', 'Velocity'});
disp(results);


%Solving numerically
h = 1;
numsteps = length(tRange);
%Using Euler's method
V_euler = zeros(numsteps, 1);
V_euler(1) = v0;
for i = 1:numsteps-1
    V_euler(i+1) = V_euler(i) + h * V_t(t(i), V_euler(i));
end
%Display results
% Display results for Euler method
relativeErrorEuler = abs((V_euler - v) ./ v);
tab_results_euler = table(t, v, V_euler, relativeErrorEuler,
    'VariableNames', {'Time', 'Actual Velocity', 'EulerVelocity',
    'RelativeError'});
disp(tab_results_euler);


%Using improved Euler (Heun) method
V_heun = zeros(numsteps, 1);

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V_heun(1) = v0;

for i = 1:numsteps-1
    slope1 = V_t(t(i), V_heun(i));
    %Predictor term
    v_predict = V_heun(i) + h * slope1;
    slope2 = V_t(t(i+1), v_predict);
    %Corrector term
    V_heun(i+1) = V_heun(i) + (h/2) * (slope1 + slope2);
end

%Display results for Heun method
% Display results for Heun method
relativeErrorHeun = abs((V_heun - v) ./ v);
tab_results_heun = table(t, v, V_heun, relativeErrorHeun, 'VariableNames',
{'Time', 'Actual Velocity', 'HeunVelocity', 'RelativeError'});
disp(tab_results_heun);

%Using RK-4 method to solve the ode
V_rk4 = zeros(numsteps, 1);
V_rk4(1) = v0;
for i = 1:numsteps-1
    k1 = V_t(t(i), V_rk4(i));
    k2 = V_t(t(i) + h/2, V_rk4(i) + h*k1/2);
    k3 = V_t(t(i) + h/2, V_rk4(i) + h*k2/2);
    k4 = V_t(t(i) + h, V_rk4(i) + h*k3);
    V_rk4(i+1) = V_rk4(i) + (h/6)*(k1 + 2*k2 + 2*k3 + k4);
end
%Display results
% Display results for RK-4 method
relativeErrorRK4 = abs((V_rk4 - v) ./ v);
tab_results_rk4 = table(t, v, V_rk4, relativeErrorRK4, 'VariableNames',
{'Time', 'Actual Velocity', 'RK4Velocity', 'RelativeError'});
disp(tab_results_rk4);

%Plot all methods on one graph
figure;
hold on;
plot(t, v, 'Marker', 'o', 'DisplayName', 'Velocity');
plot(t, V_euler, 'x-', 'DisplayName', 'Euler Method');
plot(t, V_heun, 's-', 'DisplayName', 'Heun Method');
plot(t, V_rk4, 'd-', 'DisplayName', 'RK-4 Method');
xlabel('Time (s)');
ylabel('Velocity (m/s)');
title('Comparison of Numerical Methods for Velocity');
legend('show');
grid on;
hold off;

% Solving a second-order differential equation using MATLAB (ode45)
% The equation is first converted into two first-order equations:

% dy/dt = z

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% dz/dt = (1-y^2)z - y
f = @(t,Y) [Y(2);
    (1-Y(1)^2)*Y(2)-Y(1)]; % u(1)=y, u(2)=y'

t = 0:0.2:10; %range of values
Y0 = [1;1]; % Initial conditions [y(0); z(0)] where z = y'

[t,Y] = ode45(f,t,Y0);

Y = round(Y,2);
T = table(t,Y(:,1),'VariableNames',{'t','y'});

disp(T)

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<i>Time</i>	<i>Velocity</i>
0	0
1	8.0773
2	13.492
3	17.121
4	19.554
5	21.184
6	22.277
7	23.01
8	23.501
9	23.831
10	24.051

<i>Time</i>	<i>Actual Velocity</i>	<i>EulerVelocity</i>	<i>RelativeError</i>
0	0	0	NaN
1	8.0773	9.8	0.21328
2	13.492	15.68	0.16221
3	17.121	19.208	0.12191
4	19.554	21.325	0.090585
5	21.184	22.595	0.066587
6	22.277	23.357	0.048458
7	23.01	23.814	0.034942
8	23.501	24.088	0.024985
9	23.831	24.253	0.017731
10	24.051	24.352	0.012498

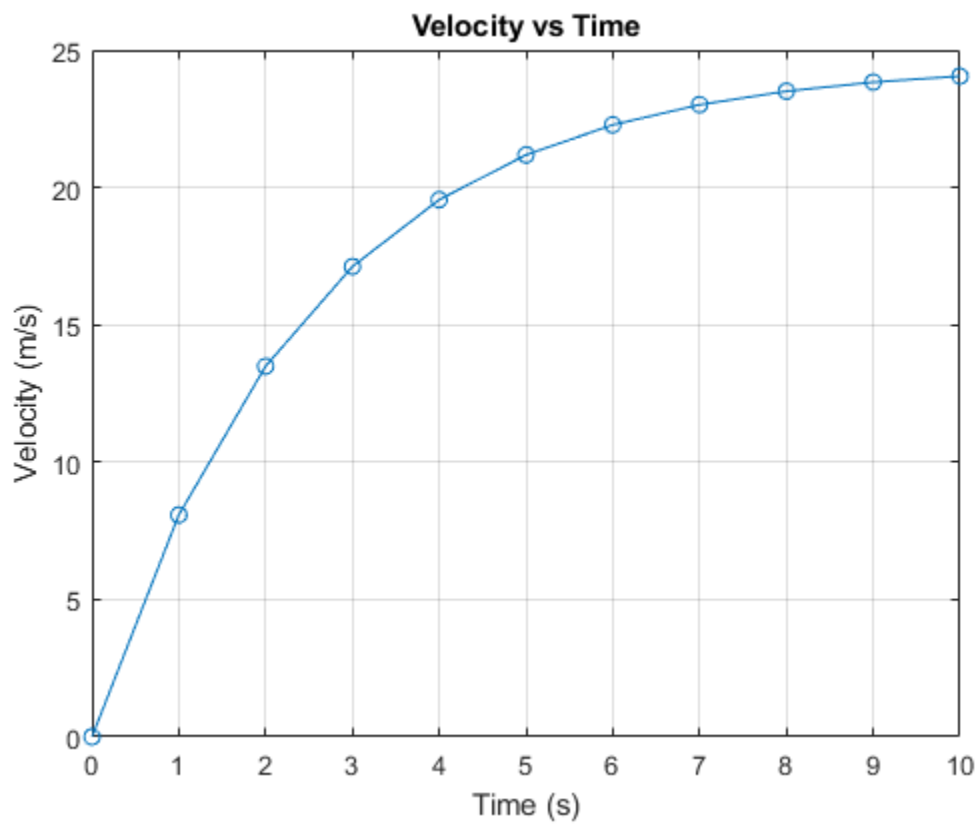
<i>Time</i>	<i>Actual Velocity</i>	<i>HeunVelocity</i>	<i>RelativeError</i>
0	0	0	NaN
1	8.0773	7.84	0.029377
2	13.492	13.171	0.023741
3	17.121	16.796	0.018945
4	19.554	19.262	0.014932
5	21.184	20.938	0.011632
6	22.277	22.078	0.0089623

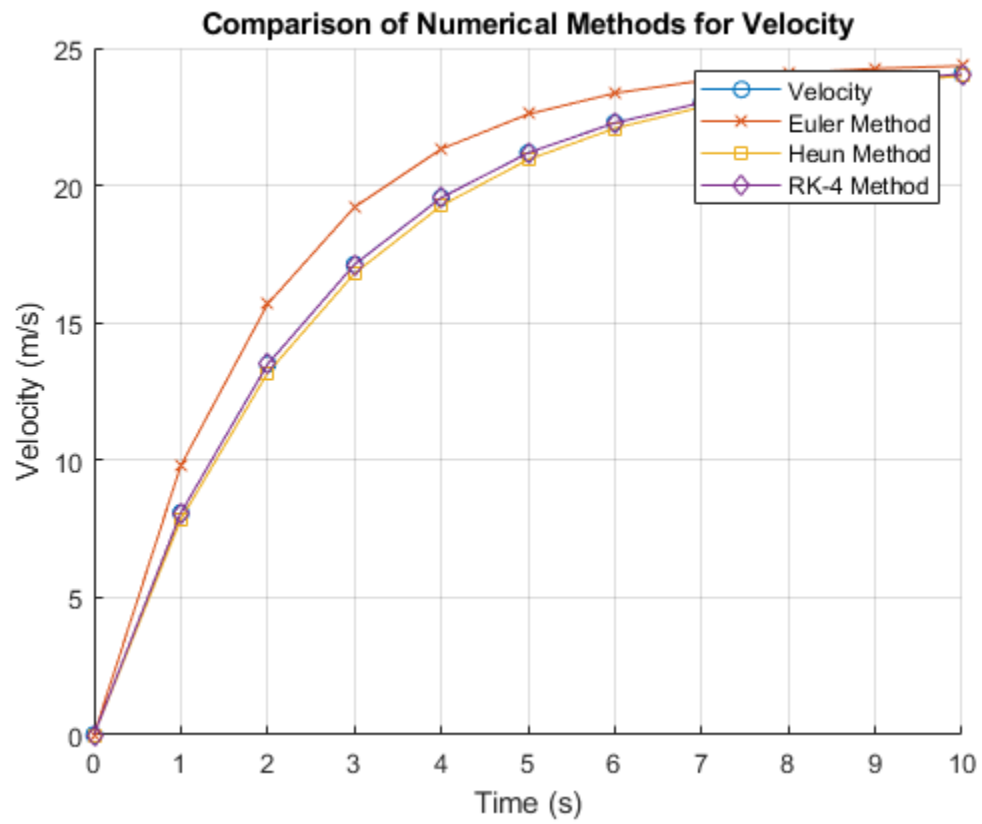
7	23.01	22.853	0.0068352
8	23.501	23.38	0.0051643
9	23.831	23.738	0.0038688
10	24.051	23.982	0.0028759

<i>Time</i>	<i>Actual Velocity</i>	<i>RK4Velocity</i>	<i>RelativeError</i>
—	—	—	—
0	0	0	NaN
1	8.0773	8.0752	0.00025844
2	13.492	13.489	0.00019929
3	17.121	17.118	0.00015576
4	19.554	19.551	0.00012106
5	21.184	21.182	9.3241e-05
6	22.277	22.276	7.1127e-05
7	23.01	23.009	5.3748e-05
8	23.501	23.5	4.0256e-05
9	23.831	23.83	2.9905e-05
10	24.051	24.051	2.1902e-05

<i>t</i>	<i>y</i>
—	—
0	1
0.2	1.18
0.4	1.3
0.6	1.35
0.8	1.35
1	1.3
1.2	1.2
1.4	1.07
1.6	0.9
1.8	0.69
2	0.42
2.2	0.08
2.4	-0.33
2.6	-0.8
2.8	-1.27
3	-1.64
3.2	-1.85
3.4	-1.93
3.6	-1.91
3.8	-1.84
4	-1.74
4.2	-1.62
4.4	-1.47
4.6	-1.3
4.8	-1.11
5	-0.88
5.2	-0.6
5.4	-0.26
5.6	0.17
5.8	0.67
6	1.19

6.2	1.63
6.4	1.9
6.6	2.01
6.8	2
7	1.94
7.2	1.84
7.4	1.73
7.6	1.59
7.8	1.43
8	1.25
8.2	1.05
8.4	0.8
8.6	0.51
8.8	0.14
9	-0.31
9.2	-0.83
9.4	-1.34
9.6	-1.74
9.8	-1.95
10	-2.01





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